

Agropioo: Actionable Intelligence for the Unconnected Farmer.

A Pakistan-First, AI-Powered Smart Agriculture Platform.



AI Hackathon
Pakistan 2026



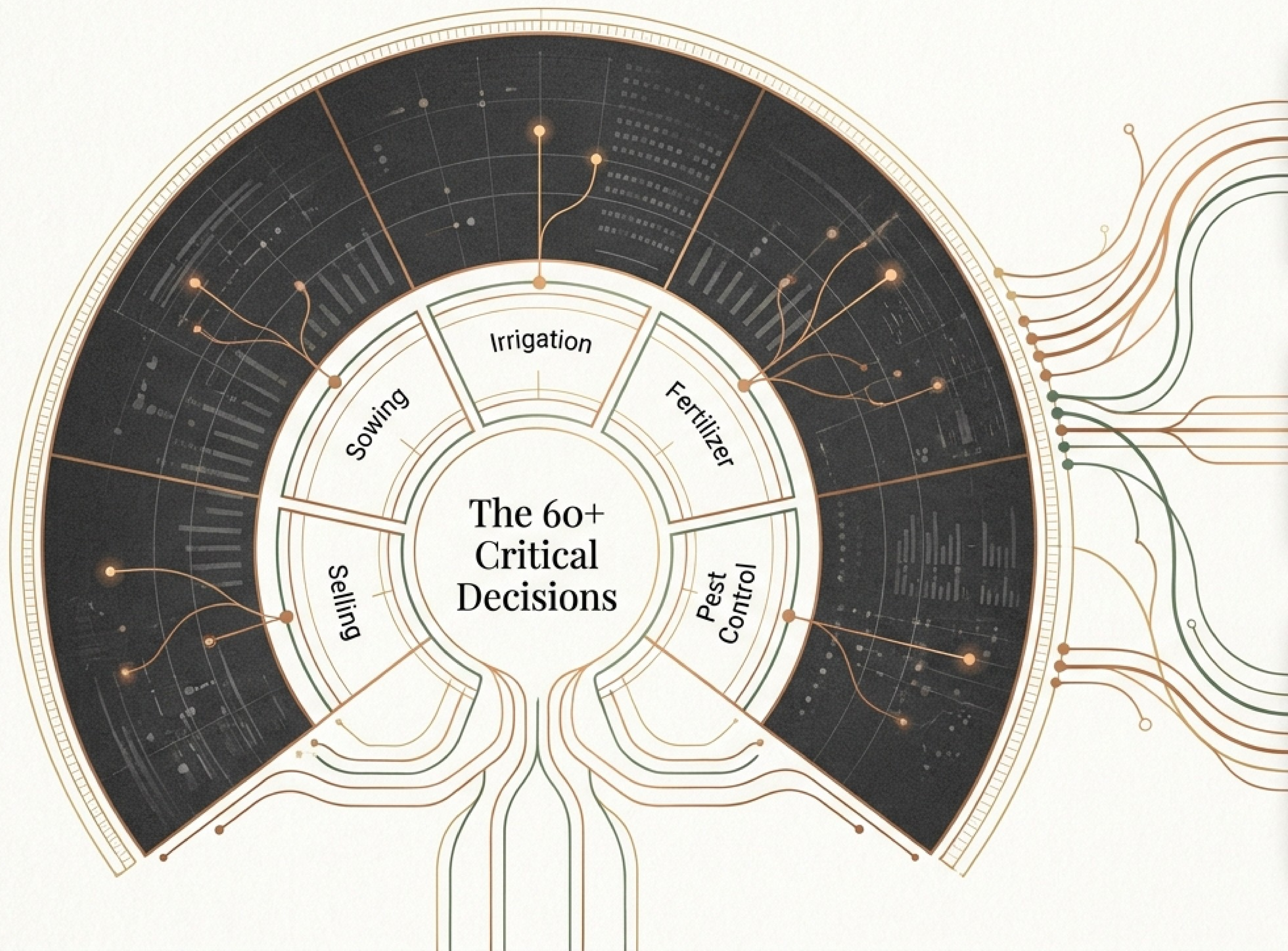
Built by Aplinode
(Sheikh Mohammad Ahmed
& Mustafa Shahzad)



Next.js 16 Native



Next.js 16 Native



The Stakes

42% of the national workforce,
23% of GDP. 45 million farmers
relying on tradition and memory.

The Penalty

30-40% of crops lost annually to
preventable disease, wasted
water, and poor timing.

The Barrier

Existing agronomic advice is
locked behind expensive physical
consultations (Rs. 2000-5000 per
visit), while extension services
reach fewer than 5% of farmers.

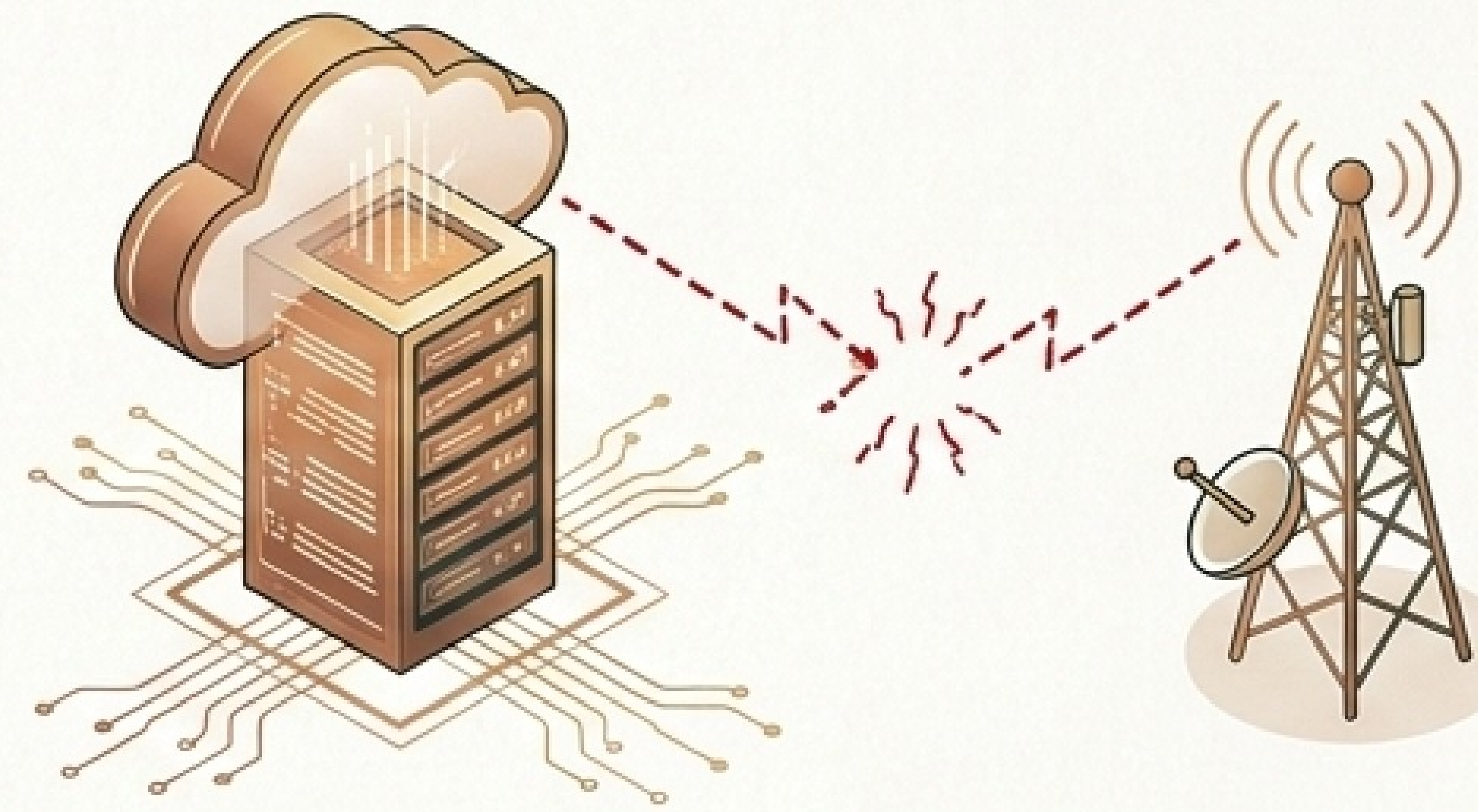
The Language Chasm



7 major regional languages

Existing apps are English/Urdu only, automatically excluding 91% of farming communities from utilizing digital tools.

The Connectivity Chasm



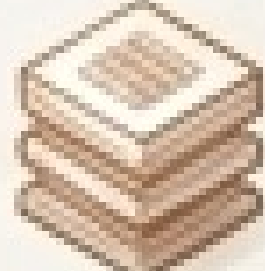
2G and intermittent network drops

Budget Android smartphones.
Cloud-dependent apps fail instantly in the field.

You cannot digitize agriculture if the tools do not speak the local language or survive the network.

AGROPIOO ARCHITECTURE

AI Advisor

- Multi-agent routing (8 distinct agents) 
- RAG-powered knowledge base 
- Conversational interface in 8 regional languages (including stylized examples of Nastaliq script)

Farm Intelligence

- Edge ML disease detection from photos
- Growth-stage-aware weather and pest alerts
- PWA offline-first syncing

Market Access

- Live mandi (market) prices scraped daily
- 14-day statistical price forecasts
- Definitive SELL/HOLD recommendations

Not just a data dashboard.
Context-aware, predictive intelligence.



Triage Agent
(Router)

Crop
Advisor

Weather

Prices

Schemes

Farm
Data

Crop
Rec

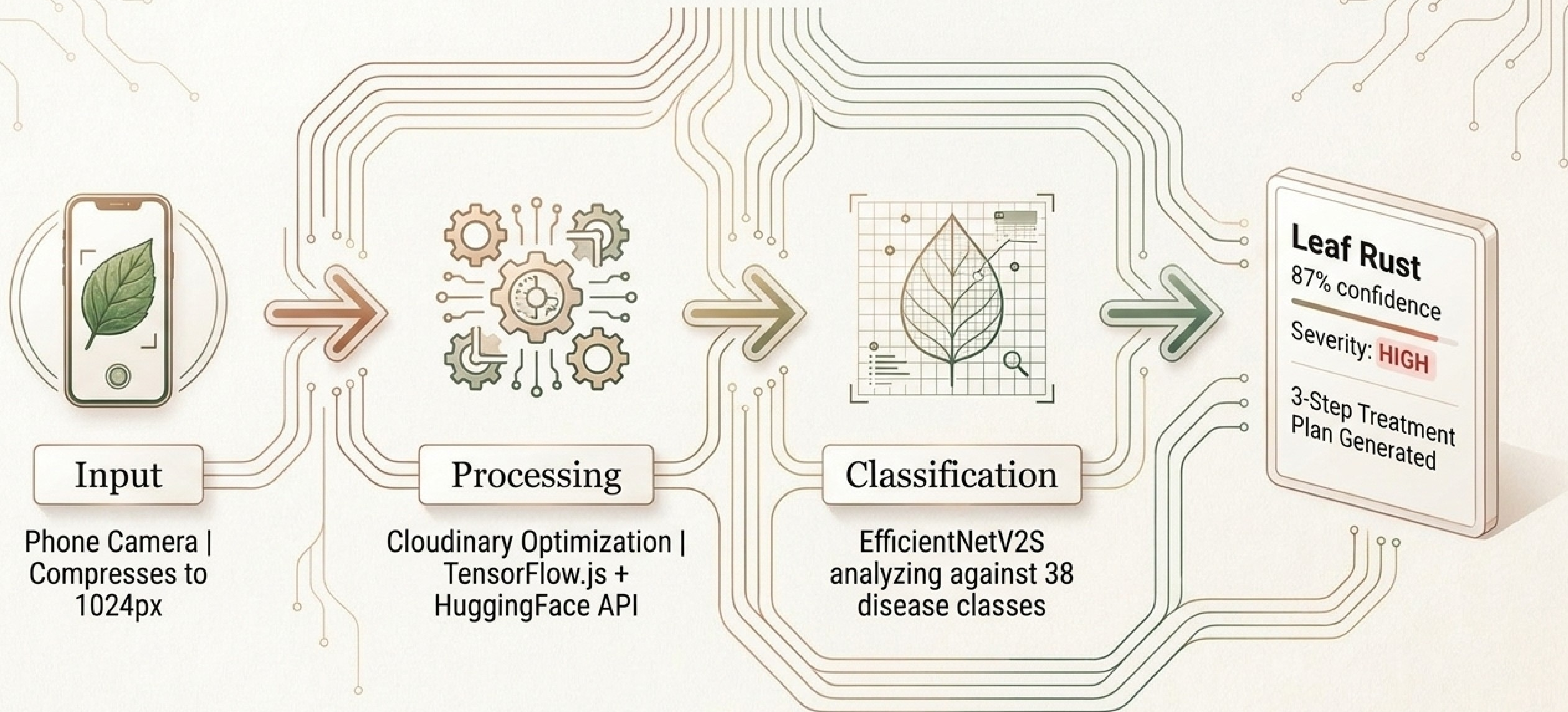
Handoff

Neon
PostgreSQL
+ pgvector

21 local RAG docs | Ollama (nomic-embed-text, 768-dim)

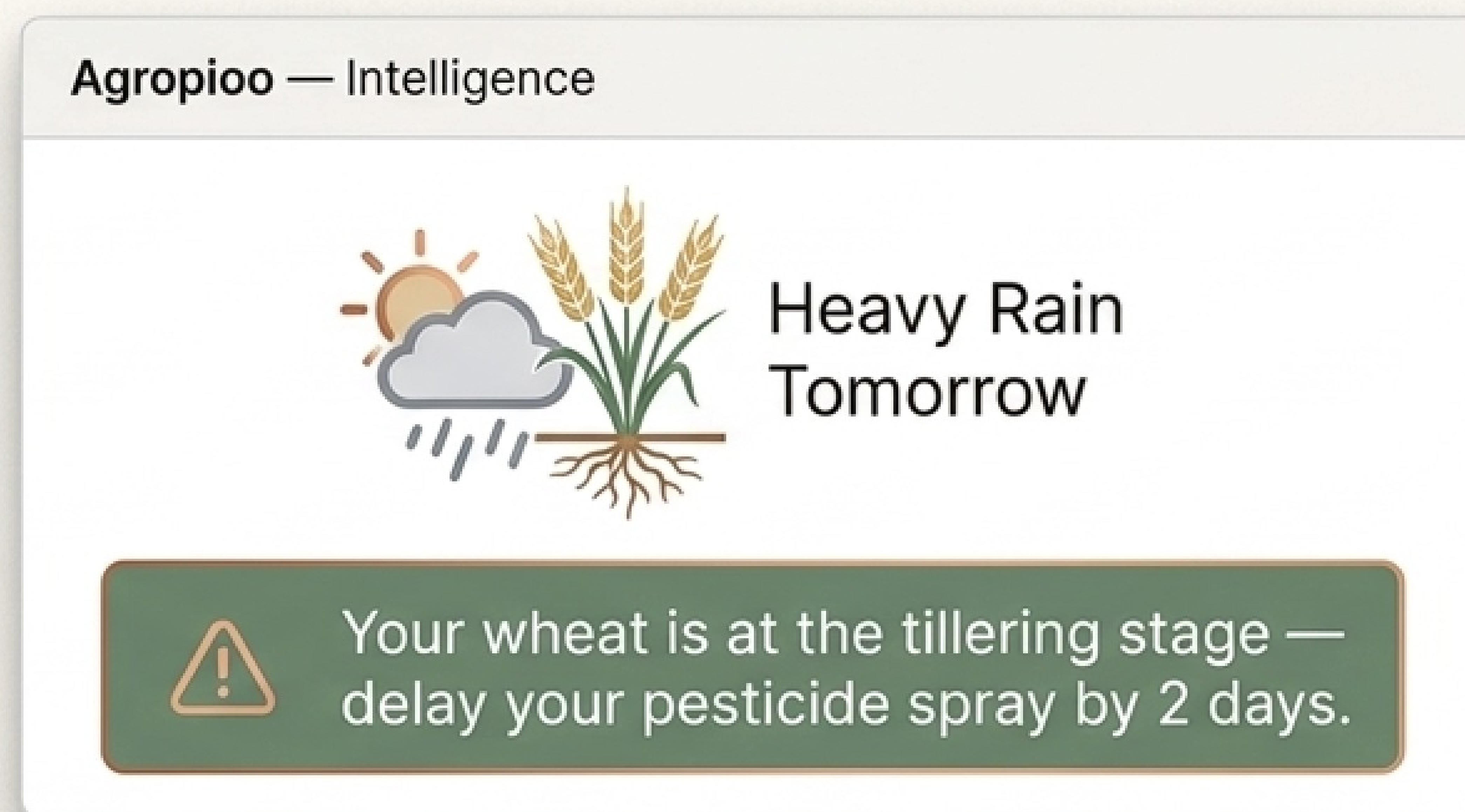
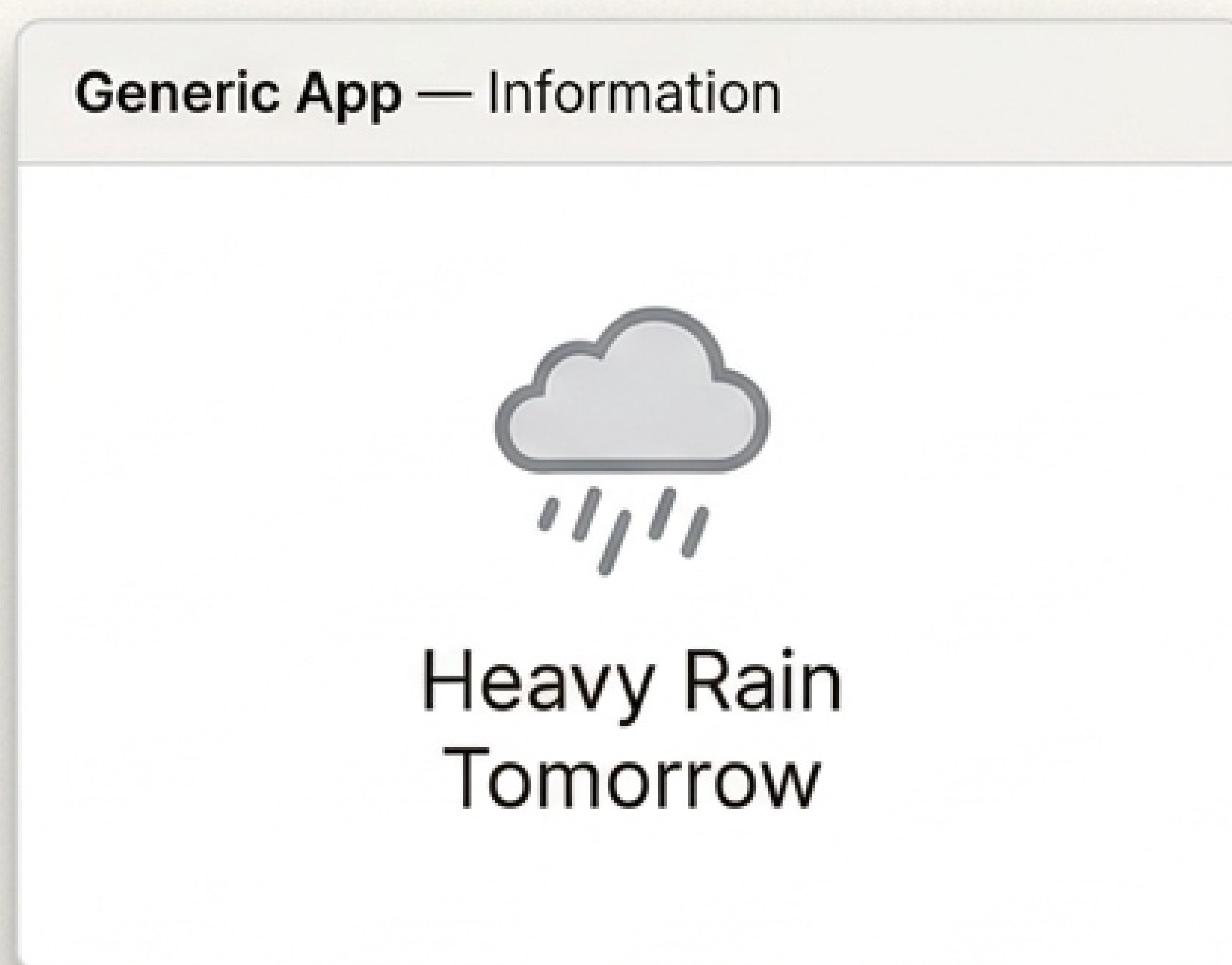
A 303-line strict farming-only
keyword guardrail ensures the
AI never deviates off-topic.

EDGE ML CROP DISEASE DETECTION FLOW



Operates offline. Photos queue in IndexedDB and process automatically when signal returns. **Zero IoT hardware** required

Generic weather is data. Growth-stage-aware advice is intelligence.



1 OpenWeather API

2 Farm Sowing Date

3 Growth-Stage
Computation

4 Cron-generated daily
pest risk scores (0-100)

Flipping Information Asymmetry

Data Ingestion

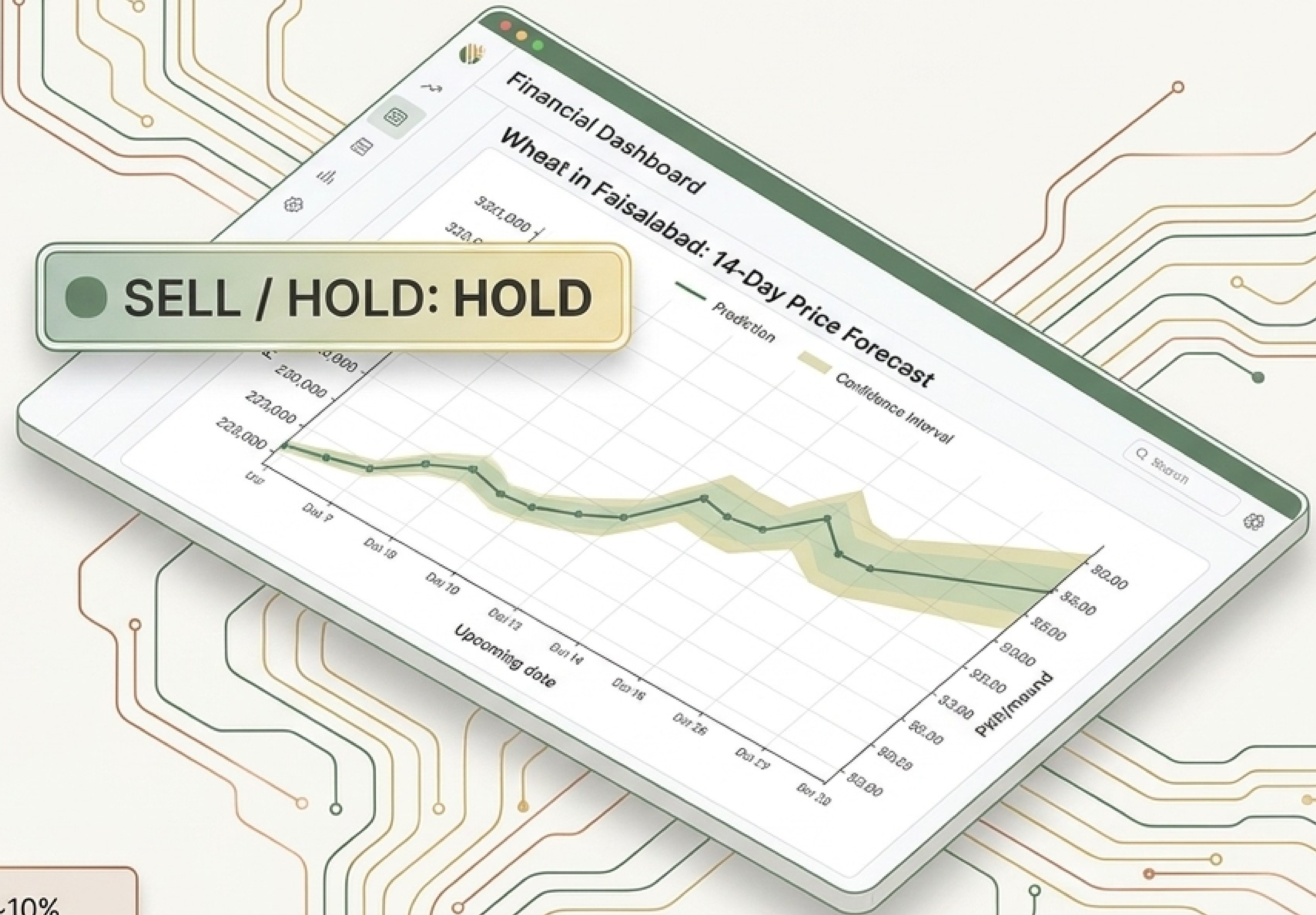
GitHub Actions cron jobs scrape 5 Gov sources daily (AMIS, SAMIS, FMIS KP, BMIS, PBS-SPI).

Scale

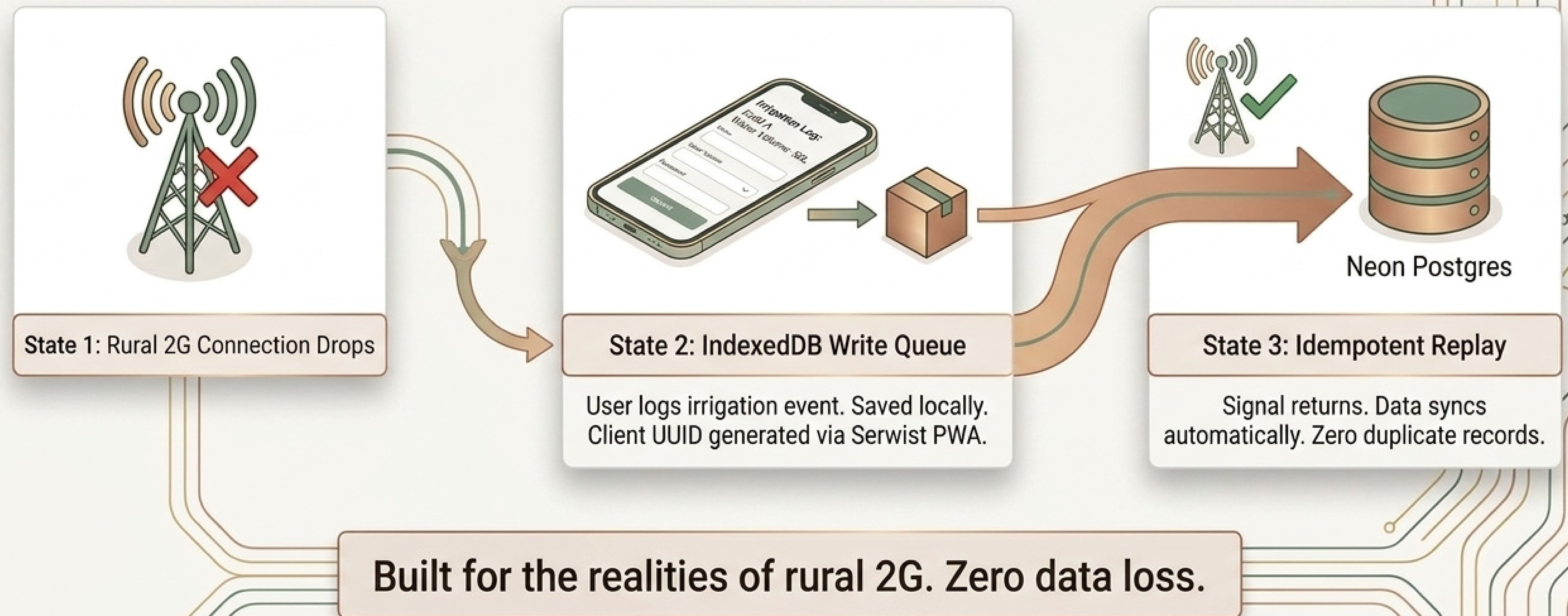
40+ crops tracked across 151 mandis (agricultural wholesale markets).

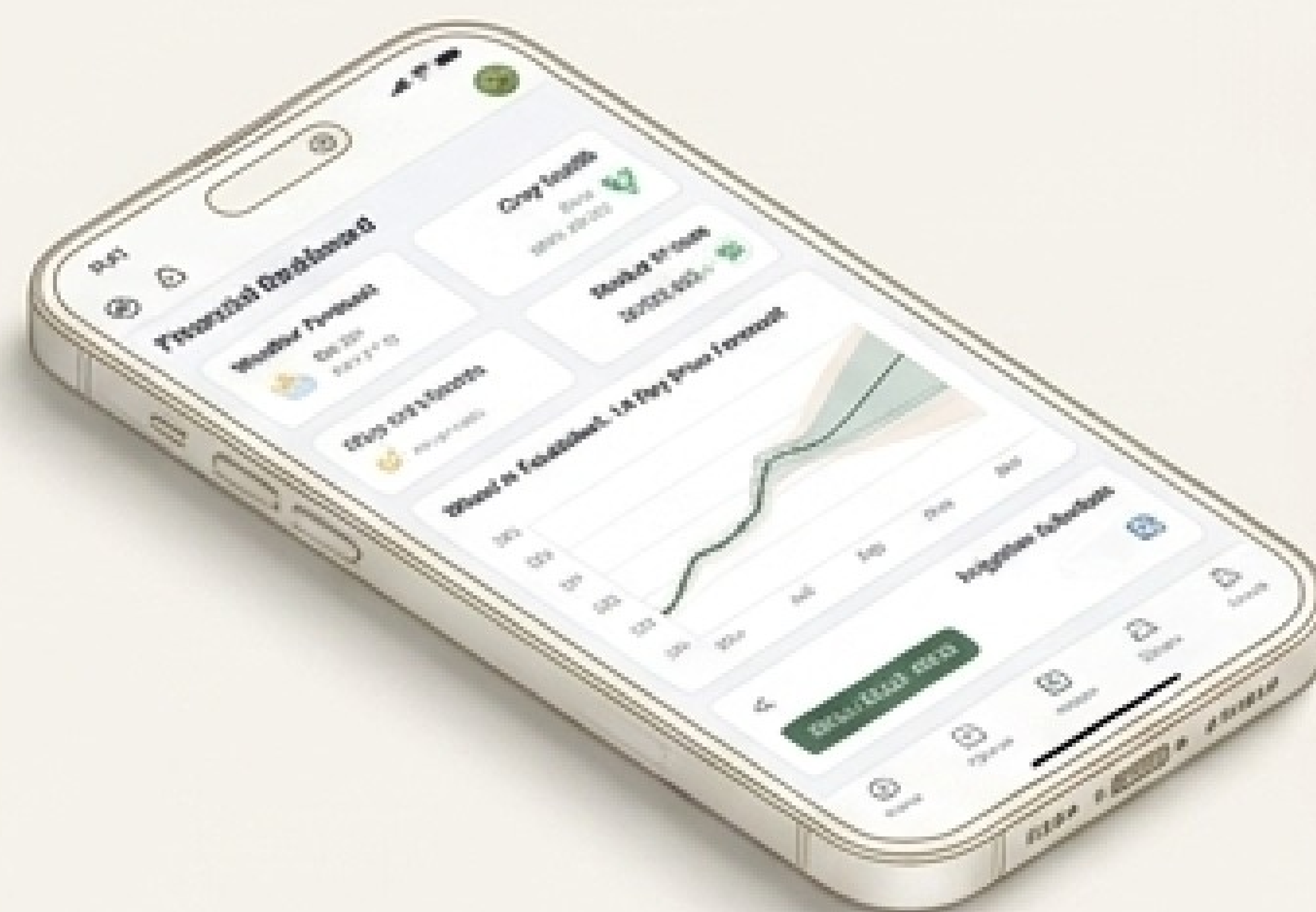
Economic Impact: Farmers traditionally lose ~10% per maund to middleman manipulation. Knowing the 14-day forecast reclaims profit margins.

● SELL / HOLD: HOLD

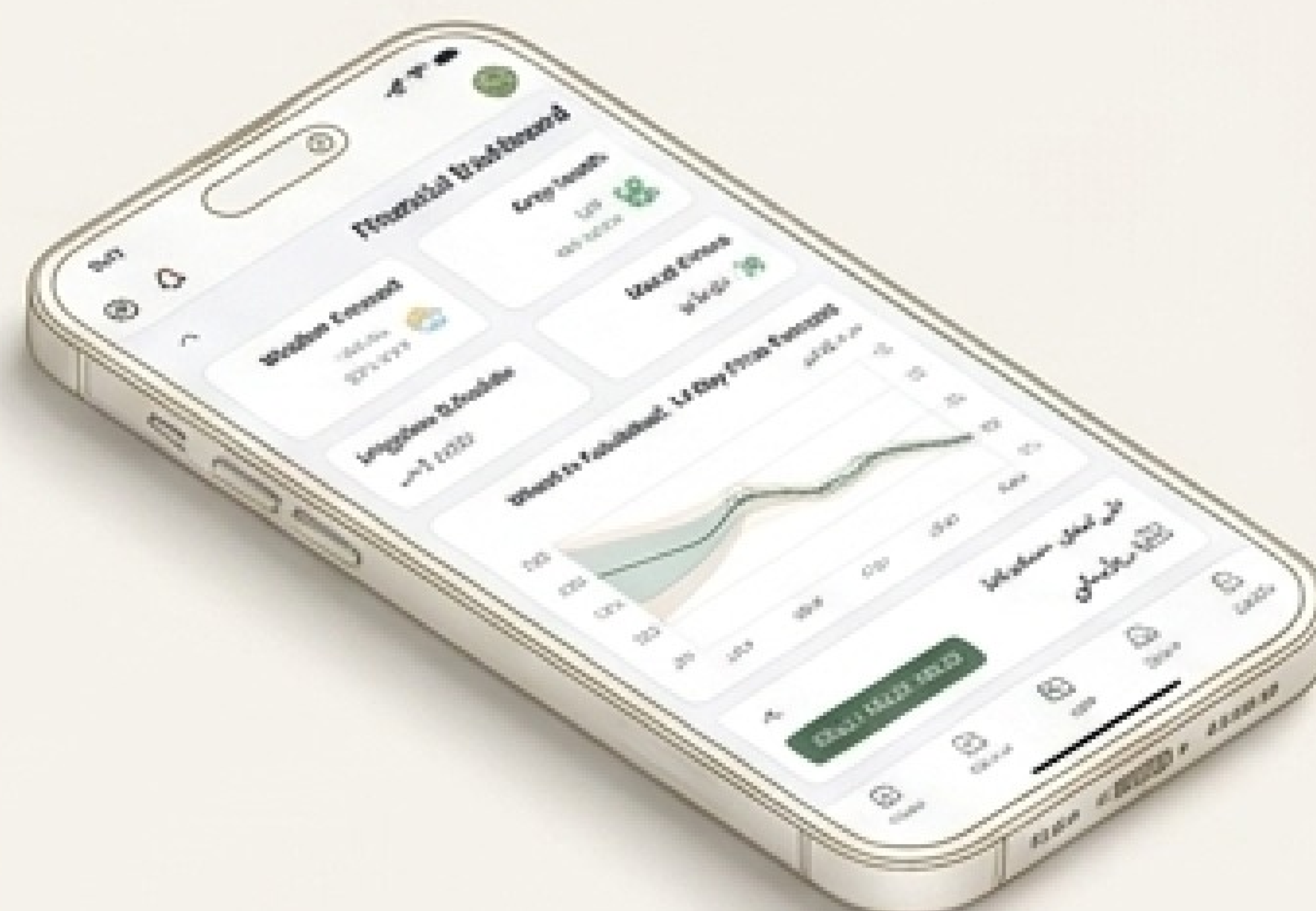


Offline-First Data Synchronization: The Digital Root System in Action





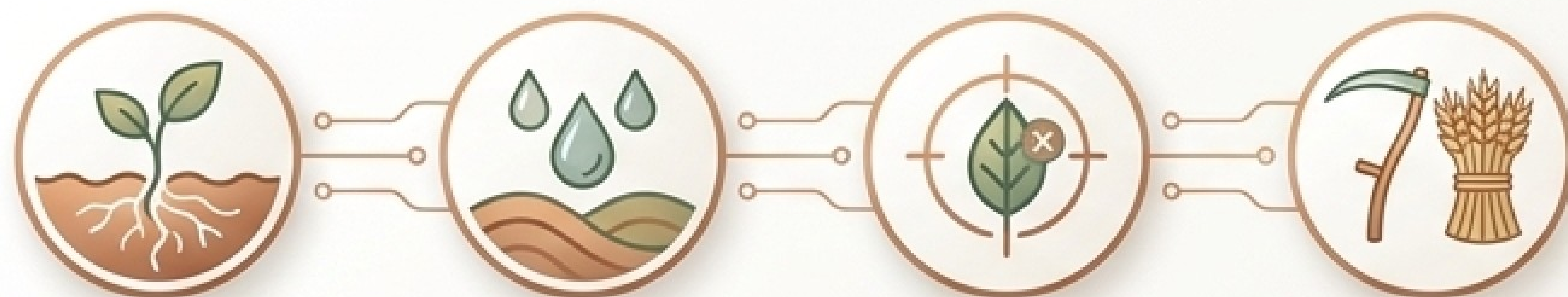
No Hardcoding: 8 locales managed completely via database-stored translations in Neon Postgres.



Native RTL: Full layout mirroring built directly into the Next.js/Tailwind v4 architecture.

Deep Routing: URL-based locale routing (/ur/dashboard).

Impact: Reaches 91% of the population.



Sowing

Irrigation

Disease

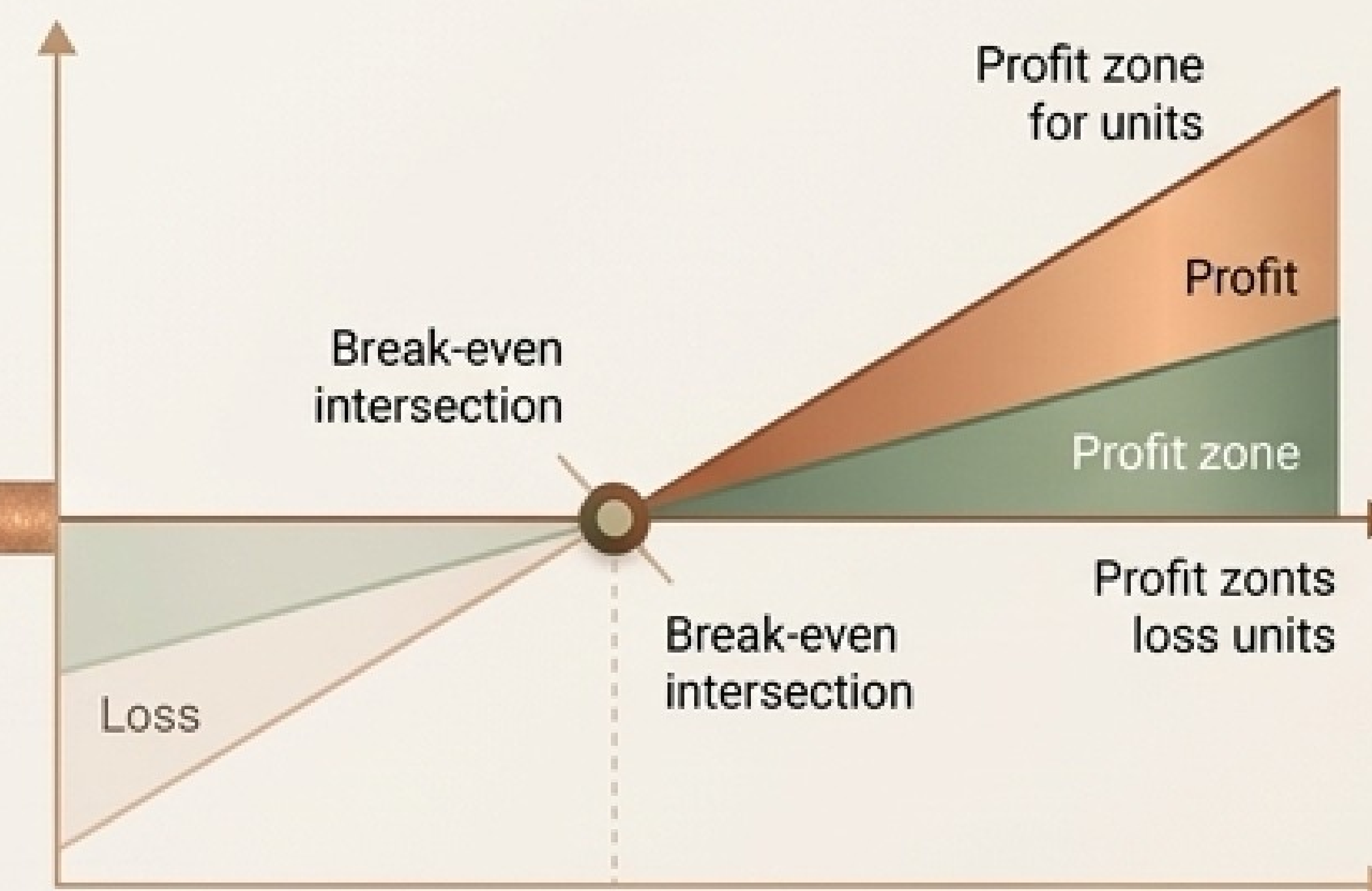
Harvest

Leaflet interactive map drawing with GPS capture & expense tracking

From zero tracking to a bank-ready financial history.

Digital ledgers allow unbanked farmers to prove income for bank loans and access modern crop insurance.

Profit & Loss Break-Even Analysis

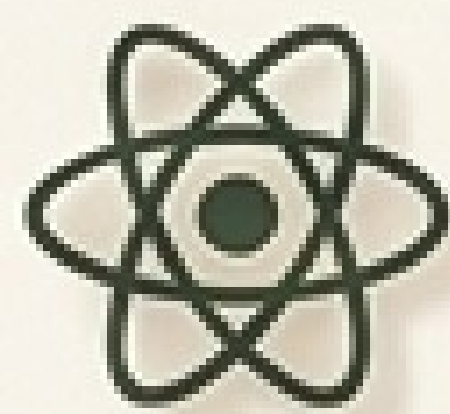


	Agropioo	Traditional Ag-Apps	Human Agronomist
Languages	8 (Native RTL)	1-2	Local
AI Architecture	Multi-Agent + RAG	Simple Chatbot	Human memory limit
Connectivity	Offline PWA Sync	Online Only	Physical Visit
Price Intelligence	14-Day Forecast (SELL/HOLD)	Current Price Only	Anecdotal
Diagnostics	Photo ML Pipeline	Text Search	Visual Inspection
Cost to Farmer	Freemium	Subscription	Rs. 2000-5000/visit

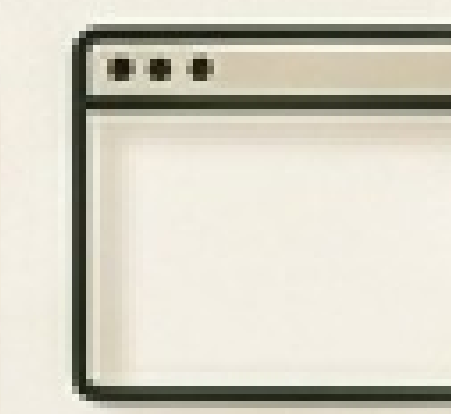
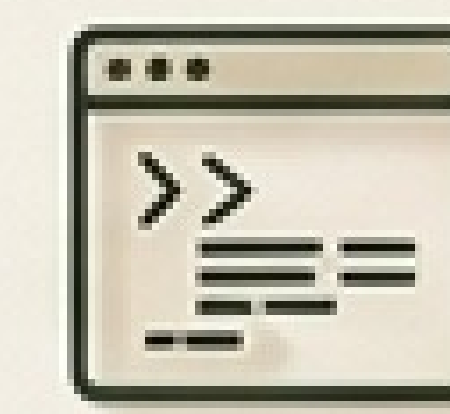
The Velocity Box

2 Builders.	2 Weeks.
16 Migrations.	66 API Routes.
85+ Components.	0 Express Backends.

Frontend Layer

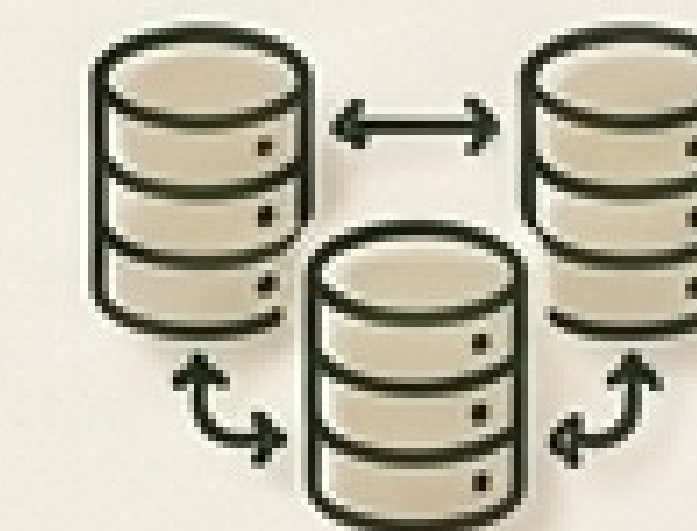


Next.js 16 | React 19 | Tailwind v4 | Serwist (PWA)



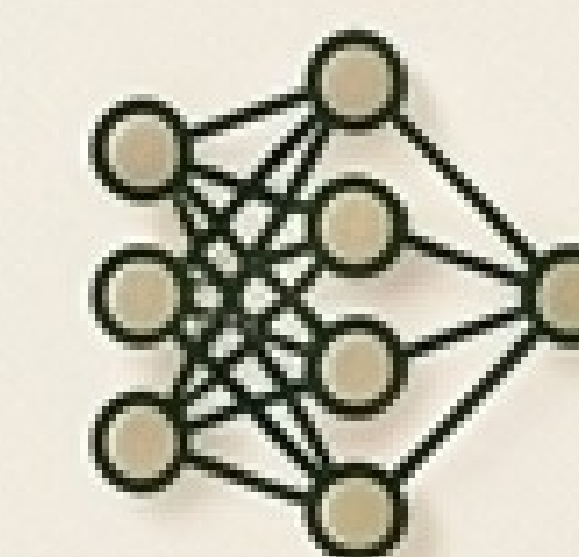
Backend/Data Layer

Neon Lakebase Postgres | pgvector | Next.js Native App Router



AI/ML Layer

@openai/agents | Ollama (Local RAG) | TensorFlow.js



STRATEGIC HORIZON

Immediate (Pilot)

Rs. 30k-80k saved per season per farmer via price timing and disease prevention.



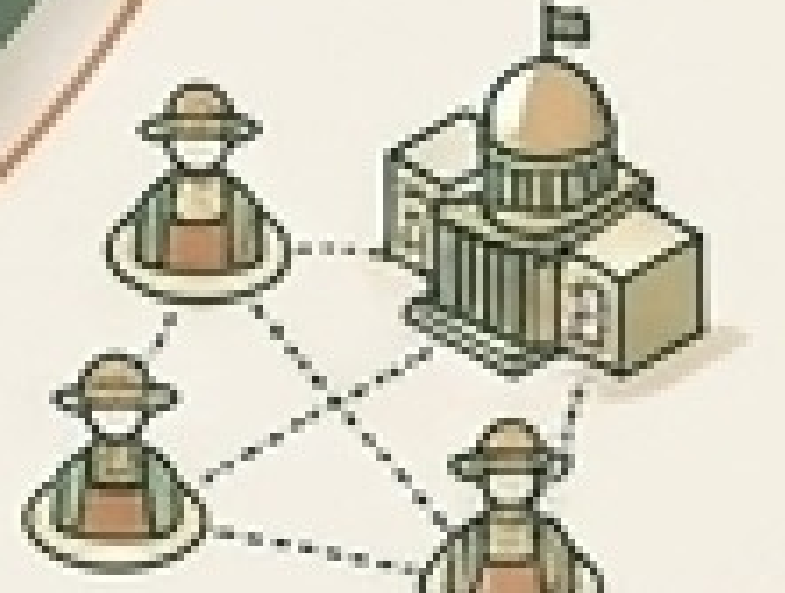
6 Months (District Scale)

Aggregate data reveals regional patterns (live pest mapping).



2 Years (National Scale)

Government subsidy integration, unlocking 40M+ unbanked farmers.



REVENUE MODEL

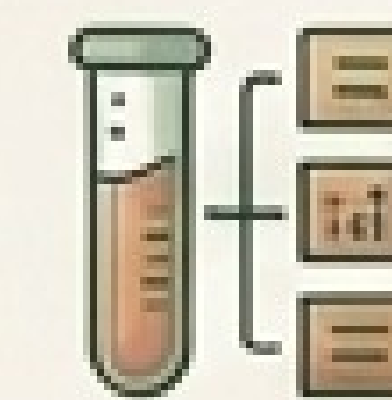
Freemium AI

Basic chat free; deep analysis premium.



B2B Agri-Tech

Anonymized aggregate data for seed/fertilizer R&D.



Insurance/Finance

Commission on enabling micro-loans via verified digital ledgers.

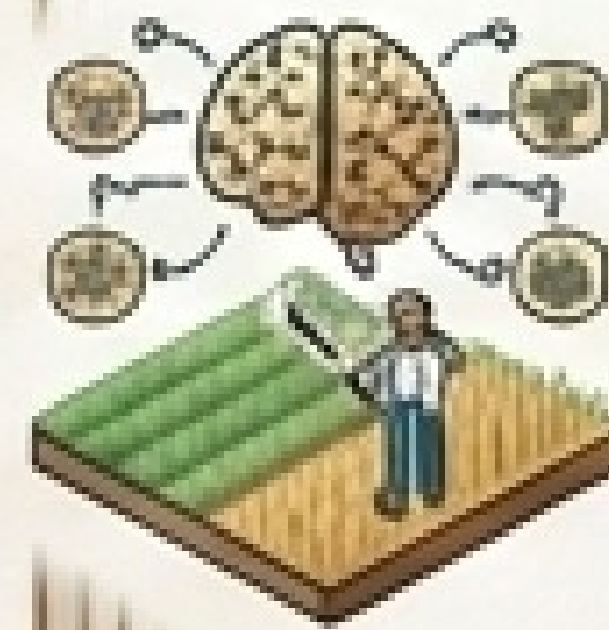


Farmer logs offline activity,
scans diseased crops, or
asks a question in Urdu



Every farmer deserves a
world-class agronomist,
in their own language,
in their own pocket.

The multi-agent AI
delivers increasingly
accurate, hyper-personalized
advice based on
growing regional data.



Data syncs via
Serwist, generating
hyper-local crop
and soil profiles.



The **pgvector** database
expands its regional
context automatically.

